

MY COMPLETE ANALYSIS REPORT (TASK 5)

1. DATA OVERVIEW

- Number of sites: 6
- Number of species: 34
- Total abundance: 121
- Mean abundance per site: 20.17
- Mean species richness per site: 13.83

2. NMDS ORDINATION (Bray-Curtis Distance)

- NMDS stress: 0
- Interpretation: EXCELLENT representation
- Number of dimensions: 2
- Convergence: 0

3. SIGNIFICANT SPECIES (envfit, $p \leq 0.05$)

- Number of significant species: 1
- Top 5 species by R^2 :
 - *S. cornuta* ($R^2 = 0.8398$, $p = 0.0333333333333333$)

4. UPGMA CLUSTERING (Average Linkage)

- Cophenetic correlation: 0.7535
- 2 clusters: 4 & 2 sites
- 3 clusters: 2, 2, 2 sites

5. PERMANOVA RESULTS (Bray-Curtis, 999 permutations)

FOR 2 CLUSTERS:

- $R^2 = 0.4088$
- $R^2 (\%) = 40.88 \%$
- p-value = 0.0666666666666667
- NO significant difference between clusters

FOR 3 CLUSTERS:

- $R^2 = 0.235$
- $R^2 (\%) = 23.5 \%$
- p-value = 0.3333333333333333
- NO significant difference among clusters

6. CONCLUSION

No significant differences in species composition were detected between the clusters identified by UPGMA.